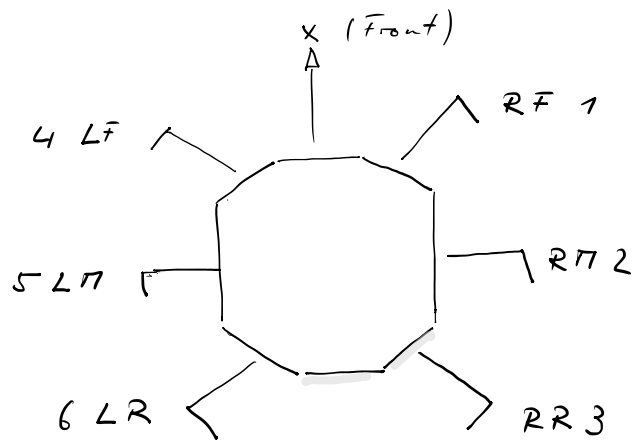
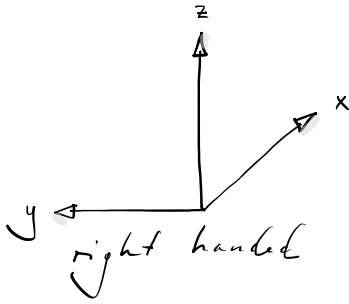
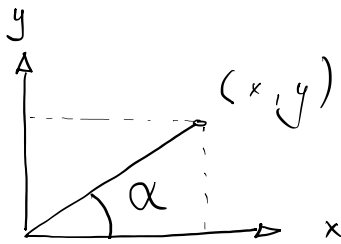


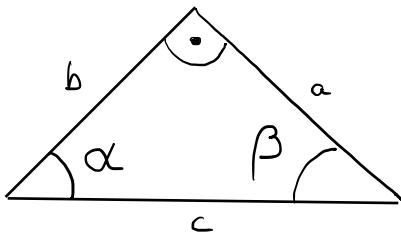
Cood System



Math Basics



$$\alpha = \arctan\left(\frac{y}{x}\right)$$

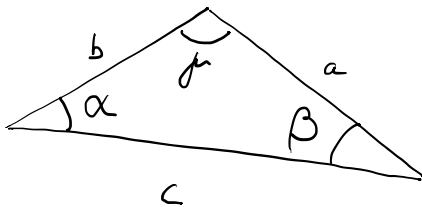


$$c = \sqrt{a^2 + b^2}$$

$$\alpha = \arctan\left(\frac{a}{b}\right)$$

$$\beta = \arctan\left(\frac{b}{a}\right)$$

$$\alpha + \beta + \gamma = 180^\circ$$

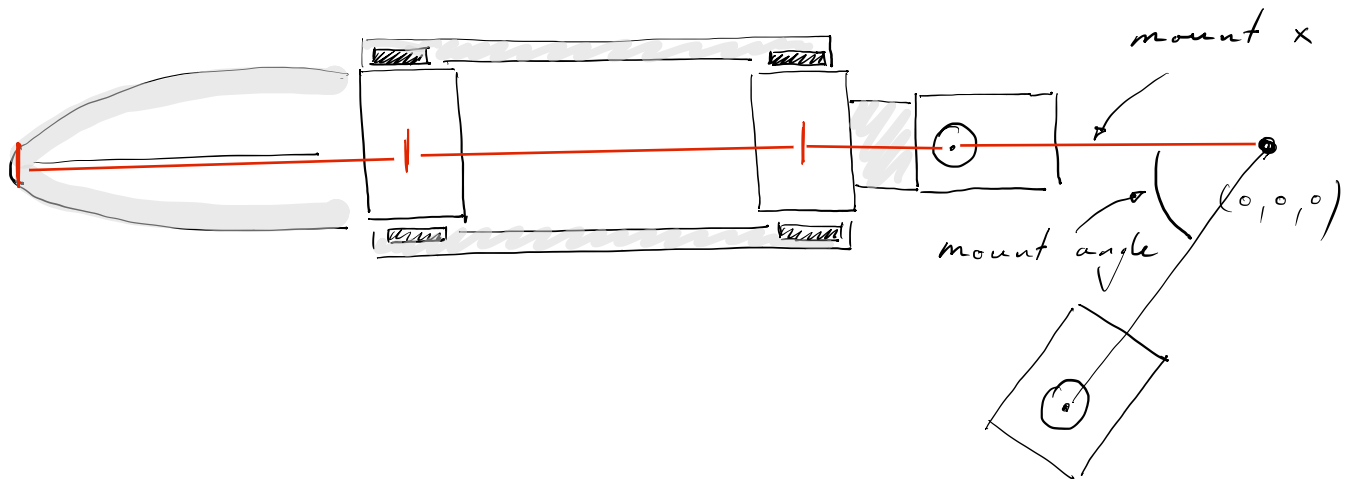
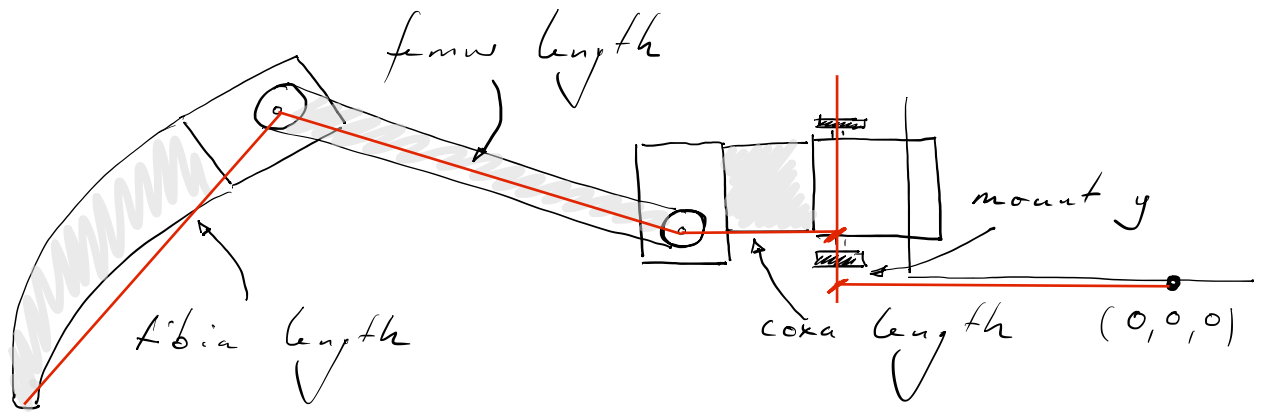


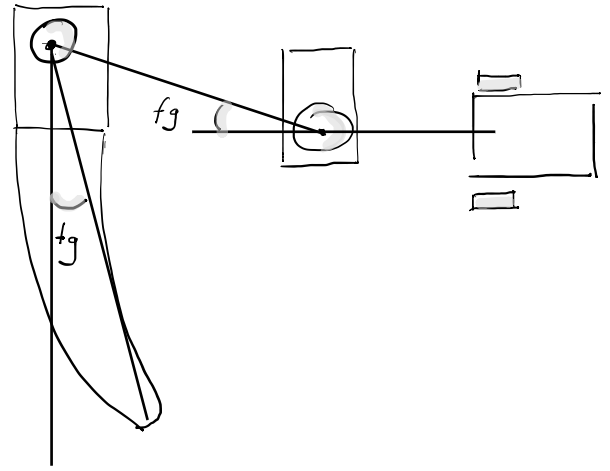
$$\alpha = \arccos\left(\frac{b^2 + c^2 - a^2}{2bc}\right)$$

$$\beta = \arccos\left(\frac{a^2 + c^2 - b^2}{2ac}\right)$$

$$\gamma = \arccos\left(\frac{a^2 + b^2 - c^2}{2ab}\right)$$

Inverse Kinematics





$$f_7 = \arccos\left(\frac{L_2^2 + L_3^2 - \ell^2}{2 \cdot L_2 \cdot L_3}\right)$$

$$f_1 = \arccos\left(\frac{L_2^2 + l^2 - L_3^2}{2 \cdot L_2 \cdot l}\right)$$

$$\textcircled{2} \quad \phi_1 = \arctan \left(\frac{p_y}{p_x} \right)$$

Go^o: coxa vero moun Lig

use h instead of y

③

include L_1 + anchor points